Fibonacci odd terms

Kind: Problem Date: 12-11-2026

1. The problem

In the Fibonacci sequence, $1, 1, 2, 3, 5, \ldots$, each term after the second is the sum of the previous two terms. How many of the first 100 terms of the Fibonacci sequence are odd?

What it asks: not the terms themselves, only a count of how many of terms 1 to 100 are odd. The 100th term has 21 digits, so listing them all is not practical.

2. The idea in one line

Odd or even depends only on the two terms before, so the sequence of odd/even is periodic with period 3: O, O, E repeating.

3. Key parts

1. **Only odd/even matters.** odd + odd = even, odd + even = odd, even + odd = odd.
2. **Find the pattern.**

Term	1	2	3	4	5	6	7	8	9
Value	1	1	2	3	5	8	13	21	34
Odd/even	O	O	E	O	O	E	O	O	E

3. **Why it must repeat.** Terms 4 and 5 are O, O, the same as terms 1 and 2. Each term's odd/even depends only on the two terms before it, so everything after term 5 repeats what came after term 2. The block O, O, E repeats for ever.
4. **Count.** Each block of 3 terms has 2 odd terms.

4. Worked solution

$100 = 3 \times 33 + 1$.

- Terms 1 to 99 are 33 full blocks, each with 2 odd terms: $33 \times 2 = 66$ odd terms.
- Term 100 is the 1st term of a new block, so it is odd: 1 more.

Number of odd terms $= 66 + 1 = \boxed{67}$.

5. Questions colleagues may ask

1. **Q. Is seeing O, O, E a few times enough?** No. A pattern seen a few times could break. The proof is that terms 4 and 5 repeat the odd/even of terms 1 and 2, and each term depends only on the two before it.
2. **Q. How many of the first 100 terms are even?** $100 - 67 = 33$, one in each full block.
3. **Q. Would the answer change if the sequence started differently?** Yes, it can. The pattern comes from the odd/even of the first two terms. For example, starting 2, 1 gives E, O, O repeating, so term 100 would be even and the count would be 66.